Section 2.4 — Factoring: Trinomials and Special Forms

Math 103: College Algebra

Factoring Quadratic Trinomials

Each quadratic trinomial below can be written as a product of two linear binomials. Factor each quadratic trinomial.

1. $3x^2 + 8x + 5$
2. $2x^2 + 9x + 9$
3. $2x^2 - x - 10$
4. $6x^2 + 5x - 6$
5. $x^2 + 12x + 35$
6. $x^2 + 5x - 14$
7. $x^2 - 8x - 20$
8. $x^2 + 8x + 12$
9. $x^2 + 3x - 40$
10. $x^2 - 11x + 28$
11. $4x^2 + 12x + 5$
12. $20x^2 + 9x + 1$
13. $5x^2 - 29x + 36$
14. $20x^2 - 39x - 80$
15. $5x^2 - 21x + 18$
16. $2x^2 + 15x - 50$

Possibly Prime Quadratic Trinomials

Factor each quadratic trinomial, if possible. If it is prime, write "prime."

17. $x^2 - 3x + 2$
18. $x^2 + 10x - 27$
19. $x^2 + 2x - 8$
20. $10x^2 + 20x + 19$
21. $5x^2 + x - 1$
22. $8x^2 + 59x - 40$
23. $x^2 + 13x - 8$
24. $20x^2 + 11x - 3$

Possible Coefficients

Using the AC method, list all nonnegative integer coefficients that can fill in the blank to allow the trinomial to be factored.

25. $x^2 - ____\, x + 20$
26. $x^2 + ____\, x - 72$
27. $x^2 - ____\, x + 5$
28. $5x^2 + ____\, x + 18$
29. $12x^2 - ____\, x - 54$
30. $4x^2 - ____\, x + 8$
31. $4x^2 - ____\, x + 54$
32. $15x^2 + ____\, x + 1$

Perfect Squares

For each trinomial below:

a. Determine whether two of its terms are perfect squares.

b. If so, determine whether the middle term makes it a perfect saure.

c. If it is a perfect square, factor it as such. (Don't factor if it isn't a perfect square.)

33. $x^2 + 12xy + 36y^2$
34. $25x^2 + 12x + 100$
35. $3x^4 + 11t^2x^2 + 8t^4$
36. $25z^2 - 60z + 36$
37. $9q^2 + 38q + 4$
38. $w^6v^2 + 10w^4v^4 + 25w^2v^6$
39. $25 - 80x + 64x^2$
40. $25x^2 + 42x + 25$

Binomial Special Forms

For each binomial below:

a. Decide which, if any, binomial special form it is.

b. If it is, factor it.

41. $x^2 - 16$
42. $5x^2 - 25$
43. $x^3 - 27$
44. $z^3 + 216$
45. $64x^3 + 1$
46. $125 - 8y^3$
47. $4x^2 - 9u^2$
48. $8 + v^6$
49. $27 - u^6$
50. $49 - u^2v^4$
51. $x^4 - 36v^4$
52. $27z^6 + v^3$
53. $49y^4 + w^2$
54. $27z^6 + 64v^3$
55. $9x^2 - 36$
56. $64 + u^4v^4$
57. $8z^6 - 125$
58. $125 + w^3$